



Fig. 2: Distributed key establishment scheme

approach to public-key cryptography is based on the mathematics of elliptic curves. ECC can obtain the same security level as RSA while using a smaller key. A 160-bit ECC key has the same security as a 1024-bit RSA key. A 224-bit ECC key compares to the 2048-bit RSA key. This is because it takes exponential algorithms to solve the elliptic-curve discrete logarithm problem as opposed to small runtime algorithms to solve the large number factorisation in RSA.

The RSA scheme generally introduces keys of 512-2048 bits. Its takes a message 'M' and composes a cipher

RSA complexity by 25 per cent.

ECC is computed by point multiplication on elliptic curves over prime integer fields or binary polynomial fields. The implementation of ECC on WSNs is primarily interested in prime integer fields since binary polynomial field mathematics is poorly supported by the slow processors. Operations of ECC scale linearly. This gives ECC an advantage over RSA on processors with small word sizes. Also, ECC grows in advantage as the key size grows.

Random key predistribution scheme. Basically, the scheme consists

of three phases: key pre-distribution, shared-key discovery and path-key establishment.

In the key pre-distribution phase, each sensor node randomly selects 'm' distinct cryptographic keys from a large key pool 'S', and stores them in its memory. This set of 'm' keys is called the node's key ring. The number of keys in the key pool, 'S', is chosen such that two random subsets of size 'm' in 'S' share at least one key with some probability 'p.'

After the nodes are deployed, a key-setup phase is performed. During this phase, each pair of neighbouring nodes attempts to find a common key that the nodes share. If such a key exists, the key is used to secure the communication link between these two nodes.

After key-setup is complete, a connected graph of secure links is formed. Nodes can then set up path keys with their neighbours with whom they do not share keys. If the graph is connected, a path can always be found from a source node to any of its neighbours. The source node can then generate a path key and send it securely via the path to the target node.

The size of key pool 'S' is critical to both the connectivity and the resilience of the scheme. Connectivity is defined as the probability that any two neighbouring nodes share one key. Resilience is defined as the fraction of the secure links that are compromised after a certain number of nodes are captured by the adversaries.

Way forward

Key management system's purpose is to secure communication in wireless sensor networks without producing much overhead. More schemes should be developed to make efficient use of sensor nodes' limited resources. Greater emphasis should be given to the security in key management schemes, particularly as a majority of sensor node deployment is in hostile environments. ●

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Threats at Different Layers in the Protocol Stack of Wireless Sensor Network

Layer	Type of attack	Attack description
Physical	Jamming	In this type of attack, adversaries interfere with the communication frequencies (radio frequencies) of the sensor nodes present in the network. For this purpose, the adversary may select a few jamming nodes from within the whole network and then apply jamming simultaneously from these selected nodes.
	Sybil attack	The base of Sybil attack is actually at the physical layer but it becomes more prompt in the higher layers like link layer and network layer. In this class of attack, the adversary introduces a malicious node into the network.
Data link	Collision	In this type of denial-of-service attack, the adversary can induce collision in only one small portion of the entire packet transmitted by a node. A small change in the data portion of the packet leads to an error in the checksum of the whole packet and asks for retransmission of the same packet.
Network	Sybil	This type of attack is prominent at the link layer.
	Black hole	In this type of attack, some of the malicious nodes in the WSN intentionally advertise zero-cost routes through them. Then some routing protocols (e.g., distance vector routing) establish a route to a destination by selecting this malicious node as an intermediate node into the routing path.
Transport	Flooding	Protocols that maintain state information at either end of the communication are vulnerable to flooding attack. One well-known attack is TCP SYN flood attack in which the adversary continuously sends the connection requests and floods the network link at the targeted node.